HKUST – Department of Computer Science and Engineering
COMP 2711: Discrete Math Tools for Computer Science
Spring 2025 Midterm Examination

Name: _____	Student ID: _____
Email: _____	Lecture: _____

Instructions

- This is a closed-book exam. You are allowed to bring one A4-sized cheat sheet, and use a calculator.
- Each question is on a separate page. This is for clarity and is not meant to imply that each question requires a full-page answer. Some can be answered using only a few lines. You may also use the back of the pages if needed.
- Please let us know if you need more scratch papers.
- Please sign the following honor code.

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I declare that the answers submitted for
this examination are my own work.

I understand that sanctions will be
imposed, if I am found to have violated the
University regulations governing academic
integrity.

Student's Name:  _____

Student's Signature:  _____
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Problems	1	2	3	4	5	6	7	8	Total
Points	15	15	12	10	12	14	8	14	100
Score									

Problem 1: Propositional logic, Inference and Proofs [15 pts]

(a) Prove that the following expression is a tautology without using the truth table

$$\neg(\neg p \rightarrow (q \wedge \neg r)) \vee \neg(\neg p \wedge r)$$

(b) Argue whether the following two expressions are equivalent or not. If they are, prove it without using truth table. If they are not, provide a counterexample.

$$\text{Expression A: } (p \wedge \neg r) \vee (\neg q \wedge r)$$

$$\text{Expression B: } (p \vee \neg q) \wedge (r \vee \neg r)$$

(c) Let p, q, r, s denote four propositions. You are given the following set of premises:

1. $\neg s \leftrightarrow (q \rightarrow r)$
2. $\neg q$
3. $(q \rightarrow s) \rightarrow (p \vee s)$

Using rules of inference for propositional logic, show that the truth of the above premises implies that the conclusion $\neg s \wedge p$ is true.

Answer: (a)

$$\begin{aligned} & \neg(\neg P \rightarrow (Q \wedge \neg R)) \vee \neg(\neg P \wedge R) \\ \equiv\ & \neg(P \vee (Q \wedge \neg R)) \vee \neg(\neg P \wedge R) \\ \equiv\ & \neg((P \vee Q) \wedge (P \vee \neg R)) \vee \neg(\neg P \wedge R) \\ \equiv\ & \neg(P \vee Q) \vee \neg(P \vee \neg R) \vee (P \vee \neg R) \equiv \neg(P \vee Q) \vee T \equiv T \end{aligned}$$

(b) They are not equivalent. $P = T, Q = T, R = T$ is a counter-example. It makes A false and makes B true.

(c) 1. $\neg q$ Premise
2. $\neg q \vee s$ From (1)
3. $(q \rightarrow s)$ Equivalent of (2)
4. $(q \rightarrow s) \rightarrow (p \vee s)$ Premise
5. $(p \vee s)$ From (3) and (4)
6. $\neg q \vee r$ From (1)
7. $q \rightarrow r$ Equivalent of (6)
8. $\neg s \leftrightarrow (q \rightarrow r)$ Premise
9. $\neg s$ From (7) and (8)
10. p From (5) and (9)
11. $p \wedge \neg s$ From (9) and (10)

Problem 2: Predicate logic [15 pts]

(a) Express the following predicates (a, b, d, m are the variables) using only $\forall$, $\exists$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$, $\neg$, $+$, $=$, $\neq$, $<$, $>$, multiplication, parentheses, variables, and constants. Take the natural numbers $\mathbb{N}$ as the domain. You are not allowed to use any other predicates.

(i) $a \equiv b \pmod{m}$

(ii) $gcd(a, b) = 1$

(iii) $gcd(a, b) = d$

(b) Express the following statements using the predicates defined above (i.e., you can use mod and gcd) and $prime(x)$, which returns true iff x is prime. You are not allowed to use other predicates. You can also use $\forall$, $\exists$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$, $\neg$, $+$, $=$, $\neq$, $<$, $>$, multiplication, parentheses, variables, and constants.

(iv) a has a multiplicative inverse modulo m if and only if a and m are co-prime.

(v) For any prime number p, every non-zero element in $\mathbb{Z}_p$ has a multiplicative inverse.

Answer: (i) $\exists r \exists q_1 \exists q_2 (r < m \wedge ra = mq_1 + r \wedge b = mq_2 + r)$

(ii) $\forall k (\exists x \exists y (kx = a \wedge ky = b) \rightarrow k = 1)$.

(iii) $\exists k (kd = a) \wedge \exists k (kd = b) \wedge \forall k (\exists x \exists y (kx = a \wedge ky = b) \rightarrow k < d)$.

(iv) $\exists b (a \cdot b \equiv 1 \pmod{m} \leftrightarrow gcd(a, m) = 1)$

(v) $\forall p (prime(p) \rightarrow \forall x (x < p \rightarrow \exists y (y < p \wedge xy \equiv 1 \pmod{p})))$

Problem 3: Peano's Axioms and Induction [12 pts]

Take the natural numbers $\mathbb{N}$ as the domain. Recall that in Peano's axiomatic system, addition is defined as follows: (i) $x+0=x$; (ii) $x+S(y)=S(x+y)$. We can further define the predicate $x \leq y$ as $\exists k(x+k=y)$. Prove that for any $x, y \in \mathbb{N}$, either $x \leq y$ or $y \leq x$ is true. You are only allowed to use Peano's axioms (including induction), definitions, and the commutative law of addition.

Answer: Proof. We will use induction on x to prove the statement.

Base case $x=0$: we will show that for any $y \in \mathbb{N}$, $0 \leq y$. By the definition of $\leq$, $0 \leq y$ means $\exists k(0+k=y)$. Taking $k=y$, we have $0+y=y+0=y$, so this is true.

Inductive step: Assume that for some $x \in \mathbb{N}$, the statement holds, i.e., for any $y \in \mathbb{N}$, either $x \leq y$ or $y \leq x$. We will prove that the statement also holds for $S(x)$, i.e., for any $y \in \mathbb{N}$, either $S(x) \leq y$ or $y \leq S(x)$.

Case 1: $y=0$: In this case, we can show that $y \leq S(x)$ following the same argument as above.

Case 2: $y \neq 0$: Since $y \neq 0$, $y=S(y')$ for some $y' \in \mathbb{N}$ (by Peano's axioms). By the inductive hypothesis, either $x \leq y'$ or $y' \leq x$.

- If $x \leq y'$, then $\exists k(x+k=y')$. Applying S to both sides gives $S(x+k)=S(y')$. By the commutative law and the definition of addition, $S(x+k)=S(k+x)=k+S(x)=S(x)+k$. This means $S(x) \leq S(y')$, i.e., $S(x) \leq y$.

- If $y' \leq x$, then $\exists k(y'+k=x)$. Applying S to both sides gives $S(y'+k)=S(x)$. By the commutative law and the definition of addition, $S(y'+k)=S(k+y')=k+S(y')=S(y')+k$. This means $S(y') \leq S(x)$, i.e., $y \leq S(x)$.

Problem 4: Cardinality [10 pts]

Determine whether each of these sets is finite, countably infinite, or uncountable. No explanation is needed.

(i) The set of all polynomials with integer coefficients, i.e., polynomials of the form $a_0 + a_1 x + \cdots + a_{k-1}x^{k-1} + a_k x^k$ for some $k \in \mathbb{N}$, where $a_i \in \mathbb{Z}$ for $i = 0, \ldots, k$.

(ii) The set of all polynomials with integer coefficients and the degree is at most 5.

(iii) The set of all polynomials with real coefficients.

(iv) The set of all polynomials with real coefficients and the degree is at most 5.

(v) The set of real roots of $1 - 2x + 3x^2 - 4x^3 + 5x^4$.

Answer: (i) Countably infinite. The union of countably many countable sets is countable. This can be proved using the same argument as showing $\mathbb{N}^2$ is countable.

(ii) Countably infinite.

(iii) Uncountable.

(iv) Uncountable.

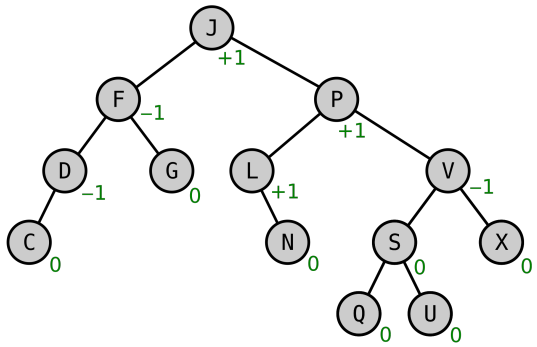
(iv) Finite.

Problem 5: Structural Induction [12 pts]

A height-balanced tree (HBT) is defined recursively as follows.

- A single node r is an HBT with height 0 and size 1.
- Two nodes u, v form an HBT, where u is the root, and v is either u's left or right child. This HBT has height 1 and size 2.
- Let T_1 and T_2 be two disjoint HBTs with roots r_1 and r_2, heights $h(T_1)$ and $h(T_2)$, sizes $n(T_1)$ and $n(T_2)$, respectively, such that $|h(T_1) - h(T_2)| \leq 1$. Create a new node r and make r_1 and r_2 the left and right child of r, respectively. The resulting structure is also an HBT with height $\max(h(T_1), h(T_2)) + 1$ and size $n(T_1) + n(T_2) + 1$.

The figure below shows an HBT T with height $h(T) = 4$ and size $n(T) = 13$. The numbers are the height differences between the left and right subtrees.



Show that $n(T) \geq f(h(T) + 2) - 1$ for any HBT T, where $f(k)$ is the k-th Fibonacci number. Recall that $f(1) = f(2) = 1$, and $f(k) = f(k-1) + f(k-2)$ for $k \geq 3$.

Answer: Base case: For HBT T which is simply a single node r, $h(T) = 0$ and $n(T) = 1$. $f(h(T) + 2) - 1 = f(2) - 1 = 0 \leq n(T)$.

Inductive step: Assume the proposition holds for some disjoint HBTs T_1 and T_2 with $|h(T_1) - h(T_2)| \leq 1$, i.e., $f(h(T_1)+2)-1 \leq n(T_1)$ and $f(h(T_2)+2) - 1 \leq n(T_2)$.

Consider HBT T rooted at r with left subtree T_1 and right subtree T_2. By definition, $h(T) = \max(h(T_1), h(T_2)) + 1$ and $n(T) = n(T_1) + n(T_2) + 1$.

Case 1: $h(T_1) = h(T_2) + 1$. Then $h(T) = h(T_1) + 1 = h(T_2) + 2$ and

$$\begin{aligned} f(h(T) + 2) - 1 &= f(h(T) + 1) + f(h(T)) - 1 \\ &= f(h(T_1) + 2) + f(h(T_2) + 2) - 1 \\ &= (f(h(T_1) + 2) - 1) + (f(h(T_2) + 2) - 1) + 1 \\ &\leq n(T_1) + n(T_2) + 1 = n(T) \end{aligned}$$

Case 2: $h(T_2) = h(T_1) + 1$. The proof is analogous to that for Case 1.

Case 3: $h(T_1) = h(T_2)$. Then $h(T) = h(T_1) + 1 = h(T_2) + 1 \geq 1$. Note that $f(h(T) + 1) = f(h(T)) + f(h(T) - 1) \geq f(h(T)) + 0 = f(h(T))$, since Fibonacchi numbers are non-negative.

$$\begin{aligned} f(h(T) + 2) - 1 &= f(h(T) + 1) + f(h(T)) - 1 \\ &\leq f(h(T) + 1) + f(h(T) + 1) - 1 \\ &= (f(h(T_1) + 2) - 1) + (f(h(T_2) + 2) - 1) + 1 \\ &\leq n(T_1) + n(T_2) + 1 = n(T) \end{aligned}$$

Hence, by structural induction, $n(T) \geq f(h(T) + 2) - 1$ for any HBT T.

Problem 6: Modular Arithmetic [14 pts]

(a) Define $2 \uparrow n$ by $2 \uparrow 0 = 1$ and $2 \uparrow n = 2^{2 \uparrow (n-1)}$. In other words,

$$2 \uparrow n = 2^{2^{\cdot^{\cdot^{\cdot^{2}}}}} \quad \overbrace{\qquad}^{n}$$

where there are n 2's in the tower.

Compute $2 \uparrow 100 \pmod{17}$.

Hint: Use Fermat's Little Theorem.

(b) Calculate $47^{37^{27}}$ mod 11?

Answer: (a) Note that 17 is a prime number. So $2^{16} \equiv 1 \pmod{17}$. So we need to calculate $2 \uparrow (99) \pmod{16}$. It is easy to see that $2 \uparrow x \equiv 0 \pmod{16}$ for $x \geq 3$. So $2 \uparrow (99) \equiv 0 \pmod{16}$.

So we can conclude that $2 \uparrow 100 \equiv 2^{16 \cdot k} \equiv (2^{16})^k \equiv 1 \pmod{17}$

(b) $47 \equiv 3 \pmod{11}$. We also know that $3^{10} \equiv 1 \pmod{11}$ (based on little Fermat theorem).

We need to calculate $37^{27} \pmod{10}$.

$37^{27} \equiv 7^{27} \equiv 7 \cdot 7^{26} \equiv 7 \cdot 49^{13} \equiv 7 \cdot 9^{13} \equiv 7 \cdot 9 \equiv 63 \equiv 3 \pmod{10}$

$$3^{37^{27}} \equiv 3^{10k+3} \equiv 27 \cdot (3^{10})^k \equiv 27 \equiv 5 \pmod{11}$$

Problem 7: Divisibility [8 pts]

Prove that for any odd natural number a, $8 \mid a^2 - 1$.

Answer: We need to prove $(a-1)(a+1)$ is divisible by 8.

Note that $a = 2k+1$ So we know that $a-1 = 2k$ and $a+1 = 2k+2$.

$(a-1)(a+1) = 2k \cdot (2k+2) = 4k^2+4k = 4 \cdot k \cdot (k+1)$. Note that $2 \mid k \cdot (k+1)$ as k and $k+1$ are two consecutive numbers. So $(a-1)(a+1) = 4 \cdot 2k' = 8k'$ and this completes the proof.

Problem 8: Congruences [14 pts]

(a) Show that for any positive integers a, b, c, d,

$$ad \equiv bd \pmod{cd}$$

if and only if

$$a \equiv b \pmod{c}.$$

(b) Solve the following congruence. When finding the inverse, you must use the extended Euclidean algorithm and show the steps.

$$28x \equiv 32 \pmod{46}$$

Answer: (a) We know that $cd \mid ad - bd$. In other words $d(a-b) = cd \cdot k$. We can divide both sides by d. So $(a-b) = c \cdot k$, which means $c \mid a - b$. This is the same as $a \equiv b \pmod{c}$.

(b) As we can see, we cannot find the inverse of 28 modulo 46 as they are not co-prime. Note that $28x \equiv 32 \pmod{46}$ is equivalent to $14x \equiv 16 \pmod{23}$. Now we calculate the inverse of 14 modulo 23 (Euclidean algorithm), which is 5.

$$70x \equiv x \equiv 80 \equiv 11 \pmod{23}$$